

A CONCRETE QUESTION ABOUT FINITE OBSERVERS

Finite Directional Records as a Concrete Observer Model

Can one localized system acquire, retain, and expose a relational orientation record while its information, support, energy cost, and gravitational footprint are controlled in one semiclassical regime?

The observer-level question

Observer rules in quantum gravity assign finite operational content to an observer subsystem, while de Sitter observer algebras make the observer itself part of the construction [1]-[4]. We ask whether a deliberately narrow physical register can make part of that abstraction auditable: a localized system that records the relative orientation of a second localized system, stores the record for a stated proper time, and permits a later relational readout.

The aim is not to identify a rotor with an observer. It is to test whether a sharply defined candidate record resource can expose which information, dynamics, localization, and gravitational inputs an observer rule actually needs.

Why direction is useful

Finite spatial reference frames and their degradation are established subjects [5]-[7]. Direction is useful here because the task can be wholly relational. If $\mathcal{O}$ is the register and $\mathcal{D}$ the target, the unknown datum is their alignment, not either orientation relative to coordinates:

$$g_{\text{rel}} = g_{\mathcal{O}}^{(-1)} g_{\mathcal{D}} \text{ in } S\mathcal{O}(3). \quad (1)$$

Performance is the Haar-prior global risk for estimating g_{rel} . This probes the full rotational orbit and distinguishes a useful record from a large but rotationally invariant Hilbert space.

The record variable

For the covariant orientation ensemble, we begin with the relative entropy of rotational asymmetry:

$$\begin{aligned} S_{\text{dir}} &:= D(\rho_{\mathcal{O}} \parallel G_{S\mathcal{O}(3)}[\rho_{\mathcal{O}}]) \\ &= S(G_{S\mathcal{O}(3)}[\rho_{\mathcal{O}}]) - S(\rho_{\mathcal{O}}). \end{aligned} \quad (2)$$

Here $G_{S\mathcal{O}(3)}$ is rotational twirling. For this task, S_{dir} is an operational rotational-asymmetry resource. It is not identified here with an effective classical record dimension, thermodynamic entropy, Hilbert-space dimension, Casimir, quantum Fisher information, or Harlow's $S_{\mathcal{O}b}$. Whether any comparison with $S_{\mathcal{O}b}$ is meaningful is the first review question.

Lifetime is operational too: after storage time T , an allowed relational readout must still attain a declared risk. This is stability of a classical record, not generic quantum phase coherence.

The proposed theorem chain

Fix a declared class $\mathcal{C}_{\text{dir}}$ of localized register-target models, states, environments, local actions, and preparation/storage/readout protocols. The intended theorem quantifies over that class:

$$\begin{aligned} R_{\text{rel}}(T) &\geq F_{\text{record}}(S_{\text{dir}}, \text{degradation}, \text{channel error}) \\ S_{\text{dir}} &\leq F_{\text{capacity}}(\text{energy}, \text{support}, \text{gravity}; \text{model data}) \\ \Rightarrow \Phi(R_{\text{rel}}(T), T, \text{support}, \text{gravity}; \text{model data}) &\geq 0. \end{aligned} \quad (3)$$

Several arrows are rigorous on named domains: global risk requires rotational asymmetry and angular-momentum resources; confined orbital matter converts angular resource into support and excitation-energy cost; and a stipulated isotropic heat channel gives a conditional finite-time degradation law.

The primary bottleneck

The common-action physical record channel is **open**. One local action must justify acquisition, storage degradation, and readout; control the error between microscopic KMS dynamics and the effective record channel; and enter the same stress-energy ledger. De Sitter master equations and controlled Markovian approximations show what must be specified, not that this bridge follows from thermality alone [10].

What gravity and the realization add

On fixed de Sitter, exact source, transmission, and electric-Weyl reconstruction maps provide a gauge-invariant route from a conserved quadrupolar stress to an exterior witness [11]. No rigorous continuum interval for the completed source presently excludes zero. Even after zero exclusion, this fixed-background witness is not a capacity bound; collapse, horizon or quantum-extremal-surface displacement, or another invariant may be the relevant constraint.

A supported massive Skymion supplies one certified local-matter candidate, against a literature where curved and gravitating de Sitter Skymions already exist [12]. It demonstrates that the supported-profile subproblem is nonempty; the observer architecture must survive rejection of its hard wall, membrane, or field model.

Why request feedback now

The project is at a fork before its most expensive bridge is built. We seek conceptual direction on the S_{dir} -to- $S_{\mathcal{O}b}$ dictionary, the gravitational quantity that should constrain a finite register, and whether a one-action semiclassical record model clarifies the observer rule without beginning from a closed-universe path integral. This is a request for criticism and redirection, not endorsement or a publication decision.

Claim boundary: $S_{\text{dir}} \neq S_{\mathcal{O}b}$; record stability is not quantum coherence; measurement back-action is not gravitational backreaction; fixed-background response is not self-consistent gravity.

Finite directional records as an observer model

Dependency map: rigorous ingredients, conditional bridges, and the theorem still to be built

Task: infer the relative $SO(3)$ alignment of two localized physical systems - never an absolute direction.



THE PURPOSE OF THIS PACKET

Where the observer model should connect to quantum gravity

These questions are intentionally upstream of the remaining interval calculation. A negative answer to any one of them would redirect the architecture while that redirection is still inexpensive.

- 1** Is S_{dir} , used here as an operational $S_0(3)$ -asymmetry resource for a covariant alignment task, meaningfully comparable to the observer entropy S_{ob} in the observer rule, or are these fundamentally different resources?
- 2** Which gauge-invariant gravitational quantity should constrain the capacity of a finite observer register in a de Sitter causal patch: collapse, horizon or quantum-extremal-surface displacement, exterior response, or something else?
- 3** Would a one-action semiclassical model deriving relational record formation, lifetime, localization, readout, and gravitational footprint clarify the observer rule, or is a closed-universe gravitational path integral essential from the outset?

Feedback target. We are not asking whether the present realization is complete. We are asking whether this is the right operational bridge to build, which quantity should replace the provisional gravitational witness if needed, and which observer-level identification should remain deliberately absent.

Auditable Status of the Directional-Record Program

[PROVED] means an analytic or computer-assisted theorem on the stated domain. **[CONDITIONAL]** means an exact implication under a stipulated effective model or source-bound floating evidence. **[OPEN]** means a missing implication, norm estimate, or physical identification. A composite claim inherits the least-complete label of its ingredients.

A. Relational orientation as the operational task

[PROVED] Let the unknown relative orientation g in $S_0(3)$ have normalized Haar prior, let an arbitrary POVM return g_{hat} , and use full-frame chordal cost $\sin^2(\theta(g_{\text{hat}}^{-1}g)/2)$. The reference representation may contain arbitrary rotation-trivial multiplicities. For every normal state and POVM, including mixed states and invariant ancillas,

$$\begin{aligned} R_{\text{ref}} &\geq 1/(16 \langle J^2 \rangle + 8), \\ R_{\text{ref}} &\geq c_{S_03} \exp[-2 S_{\text{dir}}/3], \quad c_{S_03} = 6/(e \pi^{5/3}), \\ S_{\text{dir}} &= A_{S_03}(\rho) = D(\rho \parallel G_{S_03}(\rho)). \end{aligned} \tag{A.1}$$

The first bound follows from the spin-1 fusion matrix and a discrete Hardy inequality. The second follows from relative-entropy data processing for the covariant orbit ensemble and a global rate-distortion bound. Neither proof assumes a locally unbiased estimator, Cramer-Rao regime, purity, polarization, fixed irrep, or finite-dimensional multiplicity. This is Research Theorem W3 in THEOREMS.md and docs/global_so3_reference_risk.md.

[PROVED] The datum is the alignment of two physical frames, never orientation relative to coordinates. The loss is global over the group, not a local-sensitivity proxy. For strict integer-spin support through $j=J$, $R_{\text{ref}} \geq \sin^2[\pi/(2J+3)]$. The fermionic Finkelstein-Rubinstein sector has its separately stated projective bound.

[PROVED] Here S_{dir} is only relative entropy of rotational asymmetry. Equation (A.1) gives the necessary allocation $S_{\text{dir}} \geq (3/2) \log(c_{S_03}/R_{\text{ref}})$ where positive. It does not establish accessible classical memory, sufficiency, or a relation to S_{0b} .

[CONDITIONAL] If the reference alone undergoes stipulated isotropic rotational heat diffusion with exposure $\Gamma(T) = \int_0^T \gamma(t) dt$, the spin-1 score is attenuated exactly:

$$\begin{aligned} R_{\text{ref}}(T) &\geq (3/4)[1 - \exp(-2 \Gamma)] \\ &+ \exp(-2 \Gamma)/(16 \langle J^2 \rangle + 8). \end{aligned} \tag{A.2}$$

This is a finite-time record-risk theorem for that effective channel, not preservation of arbitrary quantum coherence. **[OPEN]** No one-action protocol yet controls preparation, storage, readout, microscopic-to-effective channel error, support stress, and backreaction on one nonempty open parameter set.

B. Capacity from localization and energy

[PROVED] For finitely many spinless nonrelativistic particles of total mass M , hard support $|x_i| \leq a$, and a ground-subtracted excitation Hamiltonian satisfying the quadratic-form inequality $H_{\text{ex}} \geq \sum_i p_i^2 / (2m_i)$, every form-domain state obeys

$$\begin{aligned} \langle L^2 \rangle &\leq 2 M a^2 E_{\text{ex}}, \\ R_{\text{ref}} &\geq 1 / (32 M a^2 E_{\text{ex}} + 8). \end{aligned} \tag{B.1}$$

The result includes mixed, zero-mean, and rare-tail states with rotation-trivial internal labels. It excludes intrinsic spin, relativistic or massless fields, soft localization, negative interaction energy, and nontrivially rotating light species. A separate compactness substitution is a declared proxy, not a general-relativistic body theorem. See W3a and docs/localized_orbital_reference.md.

[PROVED] More generally, sector floors ϵ_j yield $S_{\text{dir}} \leq \beta E + \log Z_H(\beta)$ only when $Z_H(\beta) = \sum_j (2j+1)^2 \exp(-\beta \epsilon_j)$ is finite. Covariance plus finite mean energy is insufficient; a bounded invariant counterexample permits increasingly accurate frame states at uniformly bounded energy. A physical theorem must derive growing sector floors from its matter model.

[PROVED] In the hard-supported massive-Skyrme hedgehog collective family on fixed de Sitter, with support radius a and minimum support lapse $N_w > 0$, the static energy and collective inertia obey

$$\begin{aligned} I[F] &\leq 4 M[F] a^2 / (3N_w), \\ E_j &\geq \sqrt{3N_w} j(j+1)/2 / a. \end{aligned} \tag{B.2}$$

This produces a linear large-spin floor and finite integer or projective rotational partition function. It is uniform over radial relaxation inside the adiabatic collective family (W3b).

[CONDITIONAL] The Skyrmion is therefore a concrete candidate for the capacity premise, not a complete rotating-field observer. Collective-band completeness, noncollective modes, projection error, right/isospin access, and the capacity of a marked rotating wall remain uncontrolled.

C. Certified supported profile

[PROVED] At the declared fixed-background hard-wall parameters, AU.1 is an exact-rational computer-assisted local existence theorem. It proves one solution in a certified Newton ball, strict monotonicity, a negative wall slope, and finite positive rotor inertia. The source artifact is

experiments/skyrmion_newton_reduced_hessian_rounded_exact_certificate.json, SHA256
c4c95db47470392f0963266e37b491ae49a09381464f3da97c3f97bd14e74eff.

[PROVED] This is local uniqueness on fixed pure de Sitter with a prescribed Dirichlet wall. It is not global uniqueness, dynamical stability, a rotating solution, arbitrary wall dynamics, or Einstein-Skyrme backreaction.

[CONDITIONAL] It establishes that the candidate used in later calculations is more than a shooting profile; the wall remains a physical modeling input, not a harmless regulator.

D. Conserved source, master field, and electric Weyl

[PROVED] A static quadrupolar source cannot be specified by energy density alone. W3n gives the exact static even-parity bulk conservation equations and wall tractions. W3o shows that the undeformed rigid rotational Skyrmion stress fails this gate. This rejects the rigid source, not rotating Skyrmons: the centrifugal matter deformation and moving support must be included.

[PROVED] For the declared fixed-background matter-plus-moving-membrane weak form, W3z.15-W3z.16 establish a closed positive Friedrichs operator with $||A^-1|| \leq 100$, nonzero regular forcing, and a unique nonzero weak centrifugal deformation. The source artifacts have SHA256 values

4a4e3ecd48a205860de3aa045c94d2b825c0afce1c0f12f4b96254db355b85bb and

9edc2bc479534fab3f527ce535a373da1373b45085c6fbf4602f4ee9cdd32db7. Nonzero matter deformation alone does not imply nonzero exterior response.

[CONDITIONAL] Exact Hilbert-variation and shell identities, with source-bound mesh refinement, support a completed bulk-plus-moving-membrane stress at the default point. The analytic factorization is exact; closure of the selected branch is floating rather than interval certified. The safe wording is "source-bound floating conservation audit," not an unqualified certified source.

[PROVED] Once a valid conserved stress is supplied, the source, propagation, and readout maps are exact in one frozen convention. On fixed pure de Sitter, for the static $\ell=2$ normalization, regular center behavior, and no-log horizon condition,

$$\begin{aligned} A_2 \Psi &= F, & A_2 &= -d/dr[(1-r^2/R^2)d/dr] + 6/r^2, \\ F &= 8\pi G[-r^2 N \rho'/6 + r(1+4r^2/R^2)\rho/6 \\ &\quad - r p_r/2 - j + 2r p_l], & \Delta E_{rr} &= -6 \Psi(r) Y_2/r^3. \end{aligned} \tag{D.1}$$

The Friedrichs operator has a positive exact Green kernel and $A_2 \geq 6/R^2$. W3t, W3v, and W3x prove the stress-to-master identity, ideal-shell transmission, and exterior master-to-electric-Weyl reconstruction. Their artifacts have hashes 3be17612134bb6b72535db9d3c80a74acf1271a39e04eec4b1591c9bd8967887, c83536779d2ab62684218291dd99de190569f9973417d9accd2ceff426901ade, and 3025d65fe82585e584150e956de9481380ab3f4a72d26a7cacb744974838c070. Exact maps do not certify a nonzero completed input amplitude.

[CONDITIONAL] Source-bound floating calculations reduce the exterior response to one scalar amplitude and give a stable nonzero target in the declared normalization. The feasibility artifact hash is 584a22ea3ae9807dcc9da8cd6cc20274c943c52bb23c38923cbe8e6dcf986bf7. This is design evidence, not a continuum interval including profile, origin, bulk, and wall errors.

Witness boundary. If nonzero, the proposed B_W would witness non-invisibility on a fixed background. It is not the observer-capacity bound, a self-consistent geometry, or a substitute for collapse, horizon/QES displacement, or full junction data.

E. What remains in the response certificate

[PROVED] The certification architecture is exact. Outside the compact source, the fixed-background response factorizes into one amplitude $A_{\text{ext}}=J_{\text{rigid}}+B(y)$. If the symmetric weak form obeys $q(y,v)=\text{ell}(v)$ and the adjoint obeys $q(v,z)=B(v)$, arbitrary conforming trials satisfy

$$\begin{aligned} J_{\text{hat}} &= J_{\text{rigid}} + B(y_h) + R_y(z_h), \\ |A_{\text{ext}} - J_{\text{hat}}| &\leq ||R_y||_-(q^*) + ||R_z||_-(q^*). \end{aligned} \quad (\text{E.1})$$

Zero exclusion therefore requires the distance of the corrected-amplitude interval from zero to exceed the product of the full primal and adjoint dual-residual bounds. This is W3z.17; its physical premises are not yet all closed.

[PROVED] The current interval program certifies a positive-radius primal residual-square bound, an origin primal bound, exact internal conormal cancellation, the moving-wall adjoint load, and the positive-radius weak adjoint bulk load. The corresponding artifact hashes are:

- 814da74d5c21cf96b45e9967dd5b8d297d90480a46c3f3ae7fd82ba3ffaad3e7 (correlated outer primal residual),
- 7924fb7da3bb96e92fb43f68cf9311b9ac9a6077292e69474fccf5579abab504 (origin primal residual),
- 1ee67778edc45190c9b164c45bc4a76a1b9e395d172f12b5aa13241748200e2 (conormal interface),
- ee73b3527750f91bcb2ed585df3d1d58376cbe0f4ff8db47919356872a86ed42 (wall master load), and
- 3db6d390d521494d192c5df6b4bc5dfd1ee09f6d441f0bd69c3be2d18add44f5 (positive-radius adjoint bulk load).

These are certified proof components, not a response interval.

[PROVED] W3z.29 gives the current direct positive-radius-plus-wall adjoint form-dual estimate

$$\begin{aligned} \text{delta_z,partial}^2 &\leq 0.592007476516919181, \\ \text{delta_z,partial} &\leq 0.769420221021594403. \end{aligned} \quad (\text{E.2})$$

More than 99 percent of this bound comes from the adjusted bulk value coefficient, concentrated near the first positive-radius cells; the wall term is small. The exact artifact hash is 500e56b5aa36c64846100dc59a7383b2051a12c6676fcf8e6d49574f61142d0e.

[OPEN] The regular-origin master load is absent, so (E.2) is not a full delta_z . Even before that term is added, the conservative partial bound is too broad for zero exclusion. This quantified interval loss is a representation diagnostic, not a physical no-go.

[OPEN] No rigorous interval for the completed normalized electric-Weyl amplitude B_w presently excludes zero. The exact maps, favorable floating response, and unfinished physical zero-exclusion theorem are three separate facts.

F. External claim boundary

- [PROVED]** The defensible mathematical core is: global relational $S_0(3)$ risk requires asymmetry and rotational resources; named localized matter classes convert rotational capacity into energy/support cost; one hard-wall profile and its fixed-background centrifugal inverse exist; and a valid conserved quadrupolar source has exact master, transmission, and Weyl maps.
- [CONDITIONAL]** The defensible physical construction is narrower: a hard-supported Skymion is a plausible semi-classical realization, a stipulated heat channel gives an exact degradation law, and source-bound floating calculations favor a nonzero fixed-background exterior response.
- [OPEN]** The project has not identified S_{dir} with S_{0b} ; derived acquisition-through-readout from one action; proved a common parameter window; interval-certified $|B_W|>0$; completed tensorial Israel matching; solved a rotating self-gravitating Einstein-Skyrme-de Sitter geometry; or shown that exterior Weyl response caps observer information. A closed-universe path integral is outside the present claims.
- [OPEN]** Novelty, priority, venue fit, and publishability are not mathematical outputs of the repository. The separate literature matrix gives the primary-source comparison. Search absence is not used as proof of priority.

G. Reproducibility map

Status	Result	Primary repository pointer
[PROVED]	Global risk/asymmetry/Casimir bounds	THEOREMS.md W3; docs/global_so3_reference_risk.md; focused tests
[PROVED]	Confined orbital capacity	W3a; docs/localized_orbital_reference.md; focused tests
[CONDITIONAL]	Heat-capacity composition	W3a.1; experiments/universal_observer_tradeoff_certificate.json
[PROVED]	Supported collective sector floor	W3b; docs/supported_skymion_collective_spectral_floor.md
[PROVED]	Validated hard-wall profile	AU.1; exact Newton certificate listed on page 5
[PROVED]	Fixed-background source/master/Weyl maps	W3t, W3v, W3x; artifact hashes on page 6
[CONDITIONAL]	Floating completed response	W3u, W3z.17; feasibility hash on page 6
[PROVED]	Certified residual components	W3z.24-W3z.29; hashes on page 7
[OPEN]	Full Weyl zero exclusion	Needs full primal and adjoint form-dual residuals, including regular-origin adjoint load

[PROVED] The complete wording guardrail, exact artifact index, and 219-test focused verification record are in docs/harlow_review_packet/claim_evidence_ledger.md. The full source comparison and its unresolved search limits are in literature_matrix.md and source_verification_log.md.

Adversarial distinctions. Absolute orientation versus relational alignment; quantum coherence versus classical record stability; S_{dir} versus S_{0b} ; measurement back-action versus gravitational backreaction; and fixed-background response versus self-consistent gravity are audited separately.

SHORT BIBLIOGRAPHY

Conceptual Neighbors

These twelve primary sources anchor the review question. The separate literature matrix audits 33 sources across observer rules, finite reference frames, clocks, de Sitter open systems, gauge-invariant response, Skyrmons, and rigorous numerics.

1. D. Harlow, M. Usatyuk, and Y. Zhao, "Quantum mechanics and observers for gravity in a closed universe," [arXiv:2501.02359](#) (2025).
2. D. Harlow, "Observers, alpha-parameters, and the Hartle-Hawking state," [arXiv:2602.03835](#) (2026).
3. V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An Algebra of Observables for de Sitter Space," [arXiv:2206.10780](#) (2022).
4. J. De Vuyst, S. Eccles, P. A. Hoehn, and J. Kirklin, "Gravitational entropy is observer-dependent," [arXiv:2405.00114](#) (2024).
5. E. Bagan, M. Baig, and R. Munoz-Tapia, "Aligning Reference Frames Using Quantum States," [arXiv:quant-ph/0106014](#) (2001).
6. S. D. Bartlett, T. Rudolph, R. W. Spekkens, and P. S. Turner, "Degradation of a quantum reference frame," [arXiv:quant-ph/0602069](#) (2006).
7. G. Gour, I. Marvian, and R. W. Spekkens, "Measuring the quality of a quantum reference frame: the relative entropy of frameness," [arXiv:0901.0943](#) (2009).
8. B. A. Costa and Y. E. Sanchez Sanchez, "An Upper Bound for the Mass of Microscopic Clocks," [arXiv:2602.24177](#) (2026).
9. L.-Q. Chen and F. Giacomini, "Quantum Reference Fields Transformations in Linearized Quantum Gravity," [arXiv:2606.09344](#) (2026).
10. M. Fukuma, Y. Sakatani, and S. Sugishita, "Master equation for the Unruh-DeWitt detector and the universal relaxation time in de Sitter space," [arXiv:1305.0256](#) (2014).
11. H. Kodama and A. Ishibashi, "A master equation for gravitational perturbations of maximally symmetric black holes in higher dimensions," [arXiv:hep-th/0305147](#) (2003).
12. Y. Brihaye and T. Delsate, "Skyrmion and Skyrme-Black holes in de Sitter spacetime," [arXiv:hep-th/0512339](#) (2006).

Literature boundary. Existing work already covers finite observer entropy, observer-dependent gravitational algebras, degrading quantum reference frames, physical clock limits, material reference fields, de Sitter open-system dynamics, gauge-invariant response, and curved/cavity Skyrmons. The proposed distinction is their quantitative conjunction in one narrow relational $S(3)$ record model. No priority claim is made. Journal databases and terminology outside the targeted search were not exhausted.